going to try to use TruffleRuby in your application, you should have good test coverage and make sure that TruffleRuby passes your tests.

TruffleRuby can offer a substantial performance improvement. One set of benchmarks[6] suggests a 6x improvement over Ruby 3.1, and while Ruby 3.2 is somewhat faster, TruffleRuby seems worth considering for long-running processes that are particularly performance intensive and which are not dependent on those parts of Ruby that TruffleRuby doesn't handle.

JRuby

JRuby is the older and more established Java runtime version of Ruby. After several years of falling behind core Ruby, the November 2022 release (JRuby 9.4) brings JRuby to parity with Ruby 3.1, with Ruby 3.2 support expected later in 2023. There are some substantial limitations in JRuby's support as of this writing.

- Ruby's threading constructs are not completely supported, specifically Ractors and the thread scheduler aren't supported yet. You can interoperate with Java thread-safe data, however.
- The readme for this version says "Nearly all features from CRuby's NEWS file have been implemented." Again, we recommend making sure your tests continue to run if you are considering switching to JRuby.
- Rails support for most databases is incomplete.

Improvement on all these fronts is in progress, and by the time you read this, it's possible the situation has improved.

JRuby is also available via most Ruby version managers: `rbenv install jruby-9.4.0.0` will do it if you've been following along with this book's defaults. You can also get standalone download installers at https://www.jruby.org/download. You need to have a Java Standard Edition runtime installed for the standalone versions,java[7] but there's a decent chance your computer already has one installed.

With JRuby installed standalone, you can use it to run a file with `jruby <filename>`, and you can run irb with `jirb`. If you install it using `rbenv`, then the regular `ruby` and `irb` commands work. In the standalone mode, if you are running a ruby-based command line tool like `rake` the recommendation is to use `jruby -S <COMMAND>` to ensure that JRuby is actually executing the command.

Ruby gems that compile to C will not work with JRuby.

When using JRuby, you can import any Java library in your Java class path. If you add `require java` to your file, you can access Java classes via their fully qualified class names. JRuby will convert method names from Ruby-style underlines to Java-style camel case, so calling `JavaClass.do_a_thing` will reference the Java method `doAThing`. Similarly, Java getters and setters will be converted to Ruby getter and setter methods, so a `JavaClass.getFoo()` and `JavaClass.setFoo()` will be accessible as `instance.foo` and `instance.foo = value`. JRuby will convert Ruby strings, booleans, and numbers to the appropriate Java types and vice-versa. Arrays need to call the

6. https://eregon.me/blog/2022/01/06/benchmarking-cruby-mjit-yjit-jruby-truffleruby.html
7. https://www.oracle.com/java/technologies/downloads